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Quinn Booth

github.com/quinnbooth

Education

Columbia University | *The Fu Foundation School of Engineering and Applied Science*

- B.S. in Electrical Engineering, minor in Computer Science — Graduated 05/2024
  - Relevant Coursework: Applied Machine Learning, Empirical Methods of Data Science, Data Structures, Advanced Programming, Databases, Cloud Computing with AWS, User Interface Design, Computer Networks
  - GPA: 3.92 / 4.0
- Returning for M.S. in Computer Science (09/2024) — Expected Graduation 05/2025

Skills

Languages: Python, C++, C, JavaScript, TypeScript, HTML, CSS, SQL, Verilog, SystemVerilog, Java.
Technical: PyTorch, sklearn, Flask, MySQL, AWS, GCP, React, Node, REST API, GraphQL, Linux/Unix, Scripting, Git.
Leadership: Head of 3 independent research teams, mentored 40+ students in data science and machine learning.
Communication: Journalism experience with a personal mentor, fluent English and intermediate Spanish.

Research Experience

Research Assistant | Internet Real-Time Lab | Dept. of Computer Science, Columbia University — 09/2023 - Present

- Developed firmware in C++ for low power IoT devices, improving energy efficiency by >10 times. Designed and power profiled LoRa communication circuitry. Project culminates in a co-authored conference paper with Dr. Henning Schulzrinne.

Research Assistant | Creative Machines Lab | Dept. of Computer Science, Columbia University — 09/2022 - Present

- Led software/firmware design for the Stochastic Particle Robots project, using REST APIs in C++ to send instructions to a fleet of robots. Co-authored on a conference paper accepted to IEEE ReMAR 2024. Designed the 16+ robots' circuitry and PCBs.

Research Assistant | Creative Machines Lab | Dept. of Computer Science, Columbia University — 05/2022 - Present

- Managed the Artificial Metabolism project's hardware team, prototyping and using computer vision in Python to track robotic motion. Co-authored two journal/conference papers with Dr. Hod Lipson. Work featured in upcoming Bloomberg documentary.

Research Assistant | Dept. of Statistics, Columbia University — 07/2023 - 10/2023

- Implemented cutting-edge machine learning algorithms in Python, using rate-distortion-perception theory to enhance media compression with Dr. Marcel Nutz. Completed an extensive literature review (20+ papers) on modern optimization techniques.

Work Experience & Projects

Teaching Assistant | Dept. of Computer Science, Columbia University — 08/2023 - Present

- Worked with Dr. Michelle Levine on graduate-level course Empirical Methods of Data Science. Prepared live demos on deep learning and statistical data analysis, using sklearn and PyTorch. Assisted the writing and grading of assignments in Python.

STAR Scholar | Dept. of Statistics, Columbia University — 04/2023 - 07/2023

- Trained intensively in machine learning, data science, statistics and probability at the Summer Training for Aspiring Researchers program supervised by Dr. Banu Baydil.

News Editorial Intern | The Taiwan Times | Kaohsiung City, Taiwan (remote) — 05/2022 - 08/2022

- Published 12 news and opinion pieces on The Taiwan Times' official website, covering modern scientific discoveries. Cultivated strong writing and communication skills under a personal industry mentor, Hedy Heng Yang.

Tech I (Engineer) | The Escape Game | Nashville, TN — 12/2018 - 11/2019

- Produced software in C++ and designed circuitry for puzzles in escape rooms across the U.S. Managed warehouse inventory.

ML Loyalty Analysis: Predicted customer loyalty from transaction histories using XGB and LGBM in a Kaggle competition.
Spotify Podcast Research: Scraped podcast data, determining audio features suggestive of success through statistical analysis.
AWS Social Media Site: Created a React app with GPT2 integration, hosted using AWS Amplify, DynamoDB, S3 and containers.
Fractal Visualizer: Developed an interactive website, featuring various fractals such as the Mandelbrot set and Barnsley fern.